Employee Dataset Preparation (Data Cleaning, Data Analysis Report, and Quality Data Engineering)

1. Employee data preparation: Data cleaning, Data Analysis Report, and quality data preparation strategy:

In the cleaning data extraction summary of the above-provided data analysis, the required missing data in the given data comprises the real missing dimension, numerical encoding, and structural malfunction of the dataset. From the above dataset, the dataset's quality was assessed using the engineering workflow, which detailed structural refactoring and statistical validation against the preparation standards.

The primary objective of the dataset is mainly to analyse the underlying statistical distribution of the above dataset and obtain quality data engineering workflows and a machine learning pipeline utilizing Python 3. x with pandas, seaborn, NumPy, and Matplotlib libraries for cleaned and quality data metrics preparation.

Exploration and analysis:

The data pipeline begins with rigorous data analysis of exploratory data analysis within a pandas DataFrame from its CSV value. The primary exploratory interface is .info (), describe (), and head () to diagnose. The final exploration count is based on the non-null counts and the structural distribution of the above raw data.

The above raw data distribution is obtained from a 1020-employee dataset that was filtered by 12 attributes with diagnose structural refactoring and validation across the employee dataset.

The 64-bit signed integer with Boolean indicators and unstructured string objects

df. describe () df. head () df. Info ()

The baseline distribution shows the summarized data above corresponding to the data in storage distribution, null distribution, and inventory data

Raw attribute name	Raw datatype	Non null count	Null count	Null incidence
Employee id	object	1020	0	0.00%
First name	object	1020	0	0.00%
Last name	Object	1020	0	0.00%
Age	Float 64	809	211	20.69%
Depart Region	Object	1020	0	0.00%
Status	Object	1020	0	0.00%
Join date	Object	1020	0	0.00%
salary	Float 64	996	24	2.35%
email	Object	1020	0	0.00%
phone	Int 64	1020	0	0.00%
Performance score	Object	1020	0	0.00%
Remote work	bool	1020	0	0.00%

From the above data set, the missing value is present in 211 in the Age attribute raw data, with 24 missing values in the salary attribute dataset

Missing data exploration and diagnostic summary:

The missing values in the above dataset comprise statistical bias and missing values in 235 deleted data among the age and salary categories that create a statistical flaw in the raw data category. The following consequences under the missing data category are taken into consideration

Age validation via Distribution mean

The age attribute exhibited 211 missing values in the category, accounting for 20.69% of the categorical data among the 809 observed frequencies in the category. The observed frequencies lie in the discrete range of 25, 30, 35, and 40 years.

- Sample count:809 observation
- Arithmetic mean:32.48years
- Standard deviation:5.66 years
- Interquartile range: first quartile range 25.0 years and third quartile range 40.0 years
- Distribution median:30 years

Salary diagnostic via Department medians

From the diagnostic data, among the computed values of the department-wise taken among them, 24% of the values are statistically distributed across the discrete categorical range of the business domain, technical group.

The median value is taken from the risk-pertaining group among the departments that include potential variation in the category data; missing values from the various missing salaries were imputed utilizing the salary values from the different departments operating in the range.

Functional Records	Observed Records	Mean salary($)	Median salary($)	Standard deviation($)	Imputed value applied
Admin	161	85,191.92	87,358.27	20,036.82	87,358.27
Cloud Tech	141	84,923.04	86,617.37	19,686.27	86,618.37
DevOps	184	86,016.02	85,835.98	19,265.24	85,837.98
Finance	168	82,240.06	83,516.35	20,197.92	83,516.35
HR	167	85,473.64	85,518.85	21,054.96	85,518.88
Sales	175	86,896.68	86,850.37	18,988.88	86,850.37

Duplicate audit category data:

The above data was scanned with 1020 employee data, and zero duplicate records were found among the employee id.

Telephone and Date validation:

ISO Date formatting: The raw data contains the date format datetime64 to ISO 8601 format

Phone number correction: Transformation of negative integer values by calculating the zero padding to 10 digits and the absolute value of the number

Outlier calculation:

The outlier is calculated from the continuous data categories salary and age data using the interquartile range method= (Q1-1.5*IQR to Q3 -1.5*IQR), and Z-score

Variable	Min value(yrs)	Q1(25%)	Q2(50%)	Q3(75%)	Max value	IQR bounds	Outliers
age	25	25	30	35	40	22.5,42.5	0
salary	$50,047.32	$68,392.49	$85,547.82	$100,974.03	$119,975.65	$21,469.09 to$147,714.81	0

Categorical encoding:

Categorical attributes are encoded using ordinal and numerical mappings of the categories in the employee dataset.

- Status of value: Inactive (0), pending (1), active (2)
- Performance encoded: poor (0), average (1), good (2), excellent (3) • Remote work encoded: False (0), True (1)

Between before and after analysis:

Feature /metric	(Raw dataset) employee__cleaned_dataset.csv	Cleaned dataset
Row count	1020	1020
Column count	12	19
Missing values	211	0
Missing values(salary)	24	0
Phone format	Negative integers	Standard format
Date format	Mixed string	ISO 8601
Department region	Combined department region	Separated department and region
Machine level encoding	**None**	**Ordinal and label encoding included**